

Workflow labs use the variant template: Goal → Approach → Execution → Check → Report.

Learning objectives

- Describe the role of R, RStudio, Quarto, and renv in a reproducible analysis toolchain.
- Initialise a project that a collaborator can clone and run without guesswork.
- Produce a rendered Quarto document from an R chunk that draws on a pinned package environment.

Prerequisites

R, RStudio, and Quarto installed (see Get started).

Background

A statistical analysis is only as trustworthy as the environment that produced it. A result that depends on the version of a package, the operating system of the analyst, or an undocumented installation step cannot be confidently rerun by anyone else — including the analyst six months later. The job of the toolchain is to make the set of dependencies explicit and easy to restore.

R provides the language; RStudio provides an integrated environment that keeps project files, editor, console, and version control in one place; Quarto renders source documents into articles, slides, and reports while interleaving text and executable code. The last piece, renv, pins package versions to a lockfile that lives with the project. Together these give you a project that is self-contained, rebuildable on another machine, and testable against changes in its own dependencies.

Beginners often think of reproducibility as a requirement imposed by journals. The real payoff is more selfish: a colleague can pick up a project without you having to answer questions, a revision cycle does not force you to reconstruct the environment you used six months ago, and a silent upgrade of a package does not break a graph you no longer remember how to draw.

Setup

```
library(tidyverse)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Goal

Stand up a minimal reproducible R project and demonstrate the commands that keep it reproducible: version reporting, environment snapshotting, and rendering.

2. Approach

A reproducible project has four minimum ingredients: a working directory, a package lockfile, a document that can be re-rendered, and a record of the R session. In this lab we simulate a tiny dataset, draw a figure from it, and record the environment used to produce both.

```
subject = seq_len(30),
group   = rep(c("A", "B"), each = 15),
value   = c(rnorm(15, mean = 10, sd = 2),
            rnorm(15, mean = 12, sd = 2))
)
lab
```

3. Execution

```
lab |>
  ggplot(aes(group, value, fill = group)) +
  geom_boxplot(alpha = 0.6, colour = "grey30") +
  labs(x = "Group", y = "Measured value") +
  theme(legend.position = "none")
```

The chunk above is what a reader would rerun. It must be self-contained: the data are simulated inside the document, the seed is set, and only tidyverse is required.

Project anatomy

A minimal project contains:

- `my-project.Rproj` — the RStudio project file (an anchor for the working directory).
- `renv.lock` — the JSON lockfile of pinned package versions.
- `renv/` — the project’s private package library.
- An `index.qmd` or report `.qmd` — your narrative.
- A `code/` or `R/` folder — your scripts.
- A `data/` folder — ideally small and read-only; never raw CSVs you edited by hand.

renv in three commands

```
# run once per project, after creating the .Rproj
renv::init()

# after installing or upgrading packages
renv::snapshot()

# on a new machine, after cloning the repo
renv::restore()
```

The `renv::status()` command reports drift between the library and the lockfile. In a shared project, your rule of thumb is: if `status()` reports anything other than “project is synchronised with the lockfile”, do not commit.

4. Check

```
installed_versions <- tibble(
  package = c("tidyverse", "ggplot2", "dplyr"),
  version = sapply(c("tidyverse", "ggplot2", "dplyr"),
    function(p) as.character(packageVersion(p)))
)
installed_versions
```

The two outputs above — R version plus a small table of key package versions — are what a reviewer needs to rerun the figure. Everything larger (`sessionInfo()`) lives at the bottom of the document.

5. Report

A reproducible analysis is carried out in a project folder anchored by an `.Rproj` file, with dependencies pinned in `renv.lock` and narrative plus code in a Quarto document. In this lab, 30 simulated

observations were generated from two normal distributions and plotted as a boxplot; the document was rendered with `R.R.version.string` and tidyverse `packageVersion("tidyverse")`.

The important phrase is *self-contained*. A file that someone else can run by cloning the repository and typing `quarto render` is the unit of reproducibility in this course.

Common pitfalls

- Installing packages outside the project library, then wondering why `renv::restore()` on another machine produces different output.
- Editing `renv.lock` by hand. Do not. Use `snapshot()`.
- Committing `renv/library/` to git. The lockfile is what is tracked; the library is restored locally.
- Rendering a Quarto doc that reads a file outside the project directory, then sharing only the `.qmd`.

Further reading

- Quarto documentation: quarto.org.
- renv documentation: `vignette("renv")`.
- Wickham H. *R Packages*, chapter on project organisation.

Session info

See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)